

Allison Otero

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Professional Summary

Engineering student pursuing a career in aerospace, defense, and systems development, with interests in signal processing, embedded systems, AI, and systems engineering. Focused on contributing to complex, mission-critical projects that improve performance, reliability, and safety.

Education

Master of Science: Systems Engineering Admitted, Beginning August 2027

Florida Institute of Technology – Melbourne, FL

Bachelor of Science: Biomedical Engineering, Minor in Biology Expected May 2027

Florida Institute of Technology – Melbourne, FL

Certificate: Advanced Learning Algorithms July 2025

Stanford University – Stanford, CA

Certificate: Supervised Learning: Regression and Classification July 2024

Stanford University – Stanford, CA

Skills

Systems Engineering & Test: Systems analysis & design, verification & validation (V&V), technical documentation, project coordination, full system lifecycle engineering, requirements allocation and traceability, risk management, cost and timeline analysis.

Software & Analysis: Python, MATLAB, Jira, Git/GitHub, Excel, ANSYS, statistical analysis, signal processing, feature extraction, data visualization, Machine Learning (ML) model evaluation.

Embedded Systems & Troubleshooting: Arduino Uno/Nano 33 BLE Sense, Raspberry Pi, sensor interfacing, data acquisition, hardware/software integration, root-cause troubleshooting.

Design & Prototyping: 3D modeling (CAD), prototyping, waterjet, laser cutting, machine-shop tools.

Engineering Projects

Onboard Readiness and Bio-Intelligence Telemetry (ORBIT) Jan 2026 – Present

- Lead systems engineering for a four-member team developing a wearable astronaut-readiness platform.
- Manage system architecture, interface definition, integration planning, project schedule, and a \$1,400 prototype budget.
- Translate mission and user needs into system requirements, subsystem interfaces, acceptance criteria, and measurable performance targets, including **signal-to-noise ratio (SNR) >20 dB, data loss <5%, drift <2%, response time <1 second, and classification accuracy ≥90%**.
- Develop verification procedures and conduct Python-based testing of signal quality, data loss, drift, response time, and classification performance.

Relevant Experience

AI/ML Research Assistant | Florida Institute of Technology May 2026 – Aug 2026

- Develop and evaluate embedded ML workflows using Python scripting, signal processing, feature extraction, and edge inference for health-monitoring applications.
- Integrate Arduino/Raspberry Pi sensor systems; develop test cases, conduct tests, document results and design revisions, and troubleshoot data-acquisition and hardware/software issues.
- Developed and tested a **3-class** TinyML motion-recognition model using **50-sample, 6-channel** Inertial Measurement Unit (IMU) windows, achieving **96% accuracy and 5 ms inference latency**.

Robotics Controls & Manufacturing Member | Florida Institute of Technology Aug 2025 – Present

- Integrated and tested mechanical, electrical, and controls subsystems for NASA Lunabotics.
- Performed real-time system-level troubleshooting under competition deadlines, resolving critical electrical failures and delivering a fully operational rover for competition.
- Contributed to CAD, fabrication, design iteration, system integration, and technical troubleshooting for a first-year NASA Lunabotics rover team that placed **10th in the Systems Engineering Paper and 32nd overall**.

Data Analysis Intern | Defense Informatics LLC Jun 2025 – Aug 2025

- Applied Python and Excel-based **statistical analysis and predictive modeling** to identify trends and evaluate complex datasets.
- Interpreted model outputs and performance indicators, translating findings into data visualizations and concise technical summaries to support engineering decision-making.

Girls Who Code President | Florida Institute of Technology Jun 2025 – Present

- Lead Python/CAD workshops and coordinate technical events, strengthening technical presentation, leadership, and cross-functional communication skills.